

cohort_week	week_number	avg_revenue_pe	cohort_size			event_name	event_count	phase	
2020-12-13	5	0.0298175168	20078			page_view	600	BROWSING	
2020-12-13	6	0.000397566890	20078			user_engagemer	428	BROWSING	
2020-12-20	0	0.3687044307	20078			scroll	216	BROWSING	
2020-12-20	1	0.0538418396	20078			view_item	135	PRODUCT_VIEW	
2020-12-20	2	0.02091979809	20078			view_promotion	104	PRODUCT_VIEW	
2020-12-20	3	0.02327537858	20078			select_item	28	PRODUCT_VIEW	
2020-12-20	4	0.01800336511	20078			add_to_cart	16	ADD_TO_CART	
2020-12-20	5	0.008076275939	20078			begin_checkout	3	CHECKOUT	
2020-12-27	0	0.3390773324	20078			add_shipping_inf	1	CHECKOUT	
2020-12-27	1	0.05084950723	20078			purchase	0	PURCHASE	
2020-12-27	2	0.004534736078	20078						
2020-12-06	1	0.3294381702	20078						
2020-12-06	2	0.08147778688	20078						
2020-12-13	4	0.04083011967	16232						
2020-12-06	3	0.03441519114	16232						
2020-12-06	4	0.02084149774	16232						
2020-12-06	5	0.02693362785	16232						
2020-12-06	6	0.02440414692	16232						
2020-12-06	7	0.002208842495	16232						
2020-12-13	0	1.008229635	16232						
2020-12-13	1	0.1078201407	16232						
2020-12-13	2	0.04023376933	16232						
2020-12-13	3	0.03021508369	16232						
2020-11-29	0	1.319403392	16232						
2020-11-29	1	0.3634213997	17845						
2020-11-29	2	0.2432813424	17845						
2020-11-29	3	0.0480196444	17845						
2020-11-29	4	0.01241416943	17845						
2020-11-29	5	0.02214542313	17845						
2020-11-29	6	0.006093401846	17845						

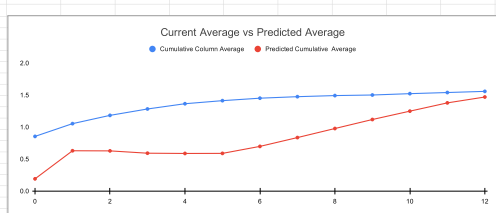
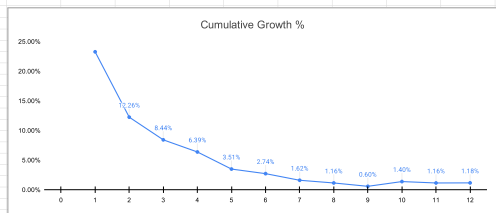
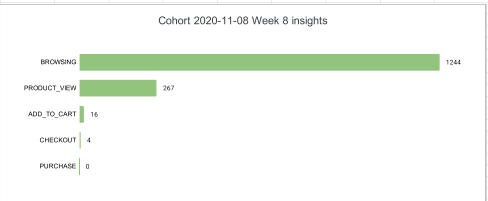
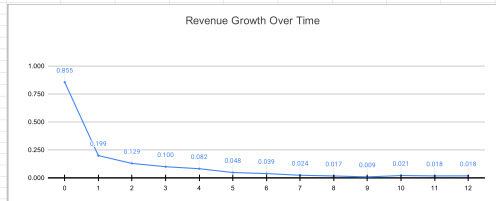
2020-11-29	7	0.01195943795	17845						
2020-11-29	8	0.005411304625	17845						
2020-12-06	0	1.202572233	17845						
2020-11-22	1	0.2358812446	17845						
2020-11-22	2	0.2253399195	17845						
2020-11-22	3	0.119366502	19637						
2020-11-22	4	0.03702194836	19637						
2020-11-22	5	0.01324031166	19637						
2020-11-22	6	0.006416458726	19637						
2020-11-22	7	0.01059224933	19637						
2020-11-22	8	0.03457758313	19637						
2020-11-22	9	0.003768396395	19637						
2020-11-15	4	0.1671056318	19637						
2020-11-15	5	0.0256094144	19637						
2020-11-15	6	0.02880358644	19637						
2020-11-15	7	0.02202297562	21991						
2020-11-15	8	0.02095825161	21991						
2020-11-15	9	0.006220229756	21991						
2020-11-15	10	0.004427010367	21991						
2020-11-22	0	1.647247543	21991						
2020-11-08	9	0.01201330705	21991						
2020-11-08	10	0.03548546082	21991						
2020-11-08	11	0.02051503204	21991						
2020-11-15	0	1.381731577	21991						
2020-11-15	1	0.2967778089	28069						
2020-11-15	2	0.2187167274	28069						
2020-11-15	3	0.2275707481	28069						
2020-11-08	2	0.2812345983	28069						
2020-11-08	3	0.2293001479	28069						
2020-11-08	4	0.2765524889	28069						
2020-11-08	5	0.104484968	28069						

2020-11-08	6	0.0393666831	28069						
2020-11-08	7	0.06936914736	25153						
2020-11-01	10	0.02315967726	25153						
2020-11-01	11	0.01494172726	25153						
2020-11-01	12	0.0181791015	25153						
2020-11-08	0	1.191966486	25153						
2020-11-08	1	0.3812838837	25153						
2020-11-01	6	0.1653053093	25153						
2020-11-01	7	0.02500249029	17830						
2020-11-01	8	0.007819503935	17830						
2020-11-01	9	0.01379619484	17830						
2020-11-01	3	0.2617292559	17830						
2020-11-01	4	0.1598764817	17830						
2020-11-01	5	0.1532025102	17830						
2020-11-01	1	0.3263771292	16539						
2020-11-01	2	0.2672078892	16539						
2020-11-01	0	0.9379918319	16539						
2020-12-27	3	0.02037608078	16539						
2020-12-27	4	0.005985851623	16539						
2021-01-03	0	0.2283744621	22774						
2021-01-03	1	0.06428383244	22774						
2021-01-03	2	0.02739966629	22774						
2021-01-03	3	0.004742249934	22774						
2021-01-10	0	0.399403319	21452						
2021-01-10	1	0.05850270371	21452						
2021-01-10	2	0.01244639194	21452						
2021-01-17	0	0.9031854489	20782						
2021-01-17	1	0.1220286787	20782						
2021-01-24	0	0.1921267894	19560						

Revenue Growth Over Time												
avg_revenue_per_user	Weeks Since User Registration											
Registration Cohort Week	0	1	2	3	4	5	6	7	8	9	10	11
2020-11-01	0.938	0.338	0.287	0.262	0.180	0.165	0.165	0.021	0.008	0.014	0.023	0.016
2020-11-08	1.192	0.381	0.281	0.229	0.277	0.104	0.030	0.069		0.012	0.035	0.018
2020-11-15	1.382	0.297	0.219	0.228	0.187	0.030	0.029	0.022	0.021	0.006	0.004	0.021
2020-11-22	1.647	0.236	0.225	0.119	0.037	0.013	0.006	0.011	0.035			
2020-11-29	1.319	0.381	0.243	0.243	0.092	0.022	0.006	0.012	0.006			
2020-12-06	1.203	0.338	0.091	0.034	0.021	0.027	0.024	0.006				
2020-12-13	1.038	0.109		0.040	0.030	0.041	0.030	0.006				
2020-12-20	0.339	0.054	0.021	0.023	0.018	0.008						
2020-12-27	0.339	0.051	0.006	0.020	0.006							
2021-01-03	0.228	0.064	0.027	0.006								
2021-01-10	0.399	0.059	0.012									
2021-01-17	0.903	0.122										
2021-01-24	0.192											
Grand Total	0.855	0.199	0.129	0.190	0.062	0.048	0.039	0.024	0.017	0.009	0.021	0.018

Cumulative Revenue Growth Over Time												
Weeks Since User Registration												
Registration Cohort Week	0	1	2	3	4	5	6	7	8	9	10	11
2020-11-01	0.938	1.264	1.532	1.793	1.953	2.108	2.272	2.297	2.305	2.318	2.341	2.356
2020-11-08	1.192	1.673	1.854	2.084	2.360	2.465	2.504	2.574	2.574	2.586	2.621	2.642
2020-11-15	1.382	1.679	1.897	2.125	2.292	2.318	2.346	2.368	2.399	2.396	2.409	
2020-11-22	1.647	1.883	2.108	2.228	2.285	2.278	2.285	2.295	2.330	2.333		
2020-11-29	1.319	1.683	1.926	1.974	1.987	2.009	2.015	2.027	2.032			
2020-12-06	1.203	1.532	1.613	1.648	1.669	1.696	1.720	1.722				
2020-12-13	1.008	1.116	1.156	1.166	1.227	1.227	1.227	1.227				
2020-12-20	0.369	0.423	0.443	0.467	0.485	0.493						
2020-12-27	0.339	0.390	0.364	0.415	0.421							
2021-01-03	0.228	0.289	0.320	0.326								
2021-01-10	0.399	0.458	0.470									
2021-01-17	0.903	1.025										
2021-01-24	0.192											
Cumulative Column Average	0.855	1.065	1.184	1.264	1.366	1.414	1.483	1.476	1.493	1.502	1.523	1.541
Cumulative Growth		23.29%	12.26%	8.44%	6.39%	3.51%	2.74%	1.62%	1.16%	0.60%	1.40%	1.16%

Predicted Cumulative Revenue Growth												
Weeks Since User Registration												
Registration Cohort Week	0	1	2	3	4	5	6	7	8	9	10	11
2020-11-01												2.335
2020-11-08											2.642	2.673
2020-11-15											2.428	2.487
2020-11-22											2.394	2.422
2020-11-29											2.097	2.122
2020-12-06											1.796	1.819
2020-12-13											1.534	1.590
2020-12-20											0.520	0.543
2020-12-27											0.463	0.480
2021-01-03											0.380	0.390
2021-01-10											0.460	0.481
2021-01-17											0.346	0.350
2021-01-24												
Predicted Cumulative Average	0.192	0.631	0.629	0.593	0.589	0.590	0.609	0.637	0.678	1.119	1.290	1.379



Key Findings						
Revenue Growth Outperformance						
- Biggest cohort attracted on week 2020-12-06 (peak acquisition week)						
- Cohorts perform strongly in week 1, then experience steep decline in value through week 12						
- Exception: Cohort 2020-11-08 had zero completions in week 8 (data gap)						
- Turning point: Week 2020-12-20 marks when overall growth started slowing						
Cumulative Growth Pattern						
- Growth drops sharply until week 9 (lowest point), then slowly recovers						
- Cohort 2020-12-13 is a key inflection point, where positive cumulative growth shifted to slower growth trajectory						
vs. Forecast						
- Current cumulative beating forecasted levels across all periods						
- Early cohorts show positive growth, then shift to smaller growth trajectory						
- Forecast model accounts for this slowdown but still outperforming						
Context Matters (Why This Happened)						
- Black Friday/Cyber Monday (2020-11-27 onwards) = inflated sales, promotions, seasonal buying						
- Christmas/New Year period typically slower for sales						
- These external factors explain both the initial spike AND the later decline						

```
WITH
registration_cohorts AS (
SELECT
user_pseudo_id,
DATE_TRUNC(MIN(PARSE_DATE('%Y%m%d', event_date)), WEEK(SUNDAY)) AS cohort_week
FROM `turing_data_analytics.raw_events`
GROUP BY user_pseudo_id
),
weekly_revenue AS (
SELECT
user_pseudo_id,
DATE_TRUNC(PARSE_DATE('%Y%m%d', event_date), WEEK(SUNDAY)) AS revenue_week,
SUM(purchase_revenue_in_usd) AS weekly_spend
FROM `turing_data_analytics.raw_events`
WHERE purchase_revenue_in_usd IS NOT NULL
GROUP BY user_pseudo_id, revenue_week
),
cohort_sizes AS (
SELECT
cohort_week,
COUNT(DISTINCT user_pseudo_id) AS cohort_size
FROM registration_cohorts
GROUP BY cohort_week
),
cohort_revenue AS (
SELECT
r.cohort_week,
DATE_DIFF(w.revenue_week, r.cohort_week, WEEK) AS week_number,
MAX(cs.cohort_size) AS cohort_size,
SUM(w.weekly_spend) AS total_revenue
FROM registration_cohorts r
INNER JOIN weekly_revenue w
ON r.user_pseudo_id = w.user_pseudo_id
AND DATE_DIFF(w.revenue_week, r.cohort_week, WEEK) BETWEEN 0 AND 12
INNER JOIN cohort_sizes cs
ON r.cohort_week = cs.cohort_week
WHERE r.cohort_week BETWEEN '2020-11-01' AND '2021-01-24'
GROUP BY r.cohort_week, week_number
)
SELECT
cohort_week,
week_number,
total_revenue / cohort_size AS avg_revenue_per_user,
cohort_size
FROM cohort_revenue
ORDER BY cohort_week, week_number;

--Validation: to check for NULL values
--SELECT 'NULL Check' as test, COUNT(*) FROM `turing_data_analytics.raw_events` WHERE event_date IS NULL OR user_pseudo_id IS NULL;

--Validation: for cohort sanity
/*
WITH reg_cohorts AS (SELECT DATE_TRUNC(MIN(PARSE_DATE('%Y%m%d', event_date)), WEEK(SUNDAY)) AS cohort_week, COUNT(DISTINCT user_pseudo_id) as size FROM `turing_data_analytics.raw_events` WHERE DATE_TRUNC(MIN(PARSE_DATE('%Y%m%d', event_date)), WEEK(SUNDAY)) BETWEEN '2020-11-01' AND '2021-01-24' GROUP BY cohort_week)
SELECT 'Cohort Distribution' as test, MIN(size) as min_cohort, MAX(size) as max_cohort, ROUND(AVG(size), 0) as avg_cohort FROM reg_cohorts;
*/

-- Validation: row count
/*
SELECT 'Raw Events' as stage, COUNT(*) as row_count, COUNT(DISTINCT user_pseudo_id) as users FROM `turing_data_analytics.raw_events`
UNION ALL SELECT 'With Purchase Revenue', COUNT(*), COUNT(DISTINCT user_pseudo_id) FROM `turing_data_analytics.raw_events` WHERE purchase_revenue_in_usd IS NOT NULL
UNION ALL SELECT 'In Cohort Date Range', COUNT(*), COUNT(DISTINCT user_pseudo_id) FROM `turing_data_analytics.raw_events` WHERE DATE_TRUNC(PARSE_DATE('%Y%m%d', event_date), WEEK(SUNDAY)) BETWEEN '2020-11-01' AND '2021-01-24';
*/
```

```
-- Segments: Best Customers, Loyal Customers, Big Spenders, Lost Customers,One-Time Purchasers, Promotion-Driven, At-Risk Customers
```

```
WITH customers_cte AS (
```

```
SELECT
```

```
CustomerID,
```

```
ANY_VALUE(Country) AS Country,
```

```
COUNT(DISTINCT InvoiceNo) AS orders,
```

```
SUM(Quantity * UnitPrice) AS order_value,
```

```
MAX(DATE(InvoiceDate)) AS last_order_date
```

```
FROM `tc-da-1.turing_data_analytics.rfm`
```

```
WHERE InvoiceDate >= '2010-12-01' AND InvoiceDate < '2011-12-02'
```

```
AND CustomerID IS NOT NULL
```

```
AND Quantity > 0
```

```
AND UnitPrice > 0
```

```
GROUP BY CustomerID
```

```
),
```

```
-- Calculate recency (days since last purchase)
```

```
rfm_value AS (
```

```
SELECT
```

```
CustomerID,
```

```
Country,
```

```
orders,
```

```
order_value,
```

```
last_order_date,
```

```
DATE_DIFF(DATE '2011-12-02', last_order_date, DAY) AS recency
```

```
FROM customers_cte
```

```
),
```

```
-- Calculate quartile thresholds
```

```
rfm_quantiles AS (
```


END AS m_score								
FROM rfm_value v								
CROSS JOIN rfm_quantiles								
),								
-- Customer Segmentation Logic								
rfm_segmented AS (								
SELECT								
CustomerID,								
Country,								
orders,								
ROUND(order_value, 2) AS order_value,								
last_order_date,								
recency,								
r_score,								
f_score,								
m_score,								
CONCAT(r_score, f_score, m_score) AS rfm_score,								
--Segments								
CASE								
-- BEST CUSTOMERS								
WHEN r_score = 4 AND f_score >= 3 AND m_score >= 3								
THEN 'Best Customers'								
-- LOYAL CUSTOMERS								
WHEN r_score = 4 AND f_score >= 3								
THEN 'Loyal Customers'								
-- BIG SPENDERS								
WHEN m_score = 4 AND f_score >= 2								
THEN 'Big Spenders'								
-- PROMOTION-DRIVEN								
WHEN r_score = 4 AND f_score >= 2 AND m_score <= 2								

THEN 'Promotion-Driven Customers'							
-- AT-RISK CUSTOMERS							
WHEN r_score <= 2 AND f_score >= 2							
THEN 'At-Risk Customers'							
-- ONE-TIME PURCHASERS							
WHEN f_score = 1 AND r_score <= 2							
THEN 'One-Time Purchasers'							
-- LOST CUSTOMERS							
WHEN r_score = 1 AND f_score = 1							
THEN 'Lost Customers'							
ELSE 'Other'							
END AS segment							
FROM rfm_scores							
)							
SELECT							
CustomerID,							
Country,							
orders,							
order_value,							
last_order_date,							
recency,							
r_score,							
f_score,							
m_score,							
rfm_score,							
segment,							
FROM rfm_segmented							
WHERE segment != 'Other'							

-- VALIDATION QUERIES									
--Check segment distribution									
/*									
SELECT segment, COUNT(*) as customer_count, ROUND(AVG(order_value), 2) as avg_value									
FROM rfm_segmented									
WHERE segment != 'Other'									
GROUP BY segment									
ORDER BY customer_count DESC;									
*/									
-- Check unclassified customers									
-- SELECT * FROM rfm_segmented WHERE segment = 'Other';									
-- Verify RFM score combinations for each segment									
/*									
SELECT segment, rfm_score, COUNT(*) as count									
FROM rfm_segmented									
WHERE segment != 'Other'									
GROUP BY segment, rfm_score									
ORDER BY segment, rfm_score;									
*/									
-- Check at-risk customers trending									
/*									
SELECT									
segment,									
COUNT(*) as count,									
AVG(recency) as avg_days_since_purchase,									
AVG(orders) as avg_lifetime_orders,									
AVG(order_value) as avg_value									

FROM rfm_segmented								
WHERE segment IN ('At-Risk Customers', 'Lost Customers')								
GROUP BY segment;								
*/								